

ERNESTO MIRÓ PERAZA

Senior Software Engineer | Full-Stack .NET/Angular | Cloud Architecture | Technical Leadership

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PROFESSIONAL SUMMARY

Computer Science Engineer with 9 years of experience analyzing, designing, developing, and evolving enterprise software solutions. Specialized in C#/ .NET and Angular, with a background in multi-tenant SaaS (Software as a Service) platforms, microservices, event-driven architecture, identity and security, and multi-cloud integration across GCP (Google Cloud Platform) and Azure. Contributed to two platform generations comprising approximately 25 services and provided technical leadership for small frontend, backend, and QA (Quality Assurance) teams. Combines hands-on development, architecture definition, and delivery coordination, while applying agentic AI (Artificial Intelligence) to improve planning, documentation, code review, and the PR (Pull Request) lifecycle. Also works on AI application capabilities using Semantic Kernel, language models, retrieval-augmented generation (RAG), and agentic workflows to connect business information and application tools.

KEY ACHIEVEMENTS

- Contributed to the evolution of approximately 25 services across two generations of multi-tenant SaaS platforms.
- Provided technical leadership for small teams and orchestrated the architecture of IGW (Itopia Gateway).
- Defined cross-cutting standards for architecture, structured logging, security, and dependency injection across 5+ repositories.
- Designed and built an end-to-end CRM (Customer Relationship Management) platform for a Mexican automotive-sector client, coordinating a 3-person team.
- Designed AI-assisted engineering workflows from backlog planning through PR creation.

TECHNICAL SKILLS

Backend & APIs: C#/ .NET (.NET Framework 4.7.2 through .NET 10), ASP.NET Core, Node.js, TypeScript, Python, REST (Representational State Transfer) APIs, gRPC (Google Remote Procedure Call), GraphQL, Apollo Server, and SSE (Server-Sent Events).

Architecture: Microservices, event-driven architecture, multi-tenant SaaS, API Gateway, state machines, sagas, callbacks, fan-out, webhooks, retry and delay queues, idempotency, and long-running asynchronous operations.

Frontend: Angular, Angular Material, RxJS (Reactive Extensions for JavaScript), Angular Signals, reactive forms, standalone components, Apollo/GraphQL with typed code generation, PWA (Progressive Web Application), MSAL.js, and WebRTC (Web Real-Time Communication) viewer integration.

Cloud & Infrastructure: GCP (Google Cloud Platform): Cloud Run, Cloud Functions, Pub/Sub, Cloud Tasks, Cloud Scheduler, Firestore, BigQuery, Cloud KMS, Secret Manager, Compute Engine, and Cloud Build. Azure: Azure Virtual Desktop, ARM (Azure Resource Manager), Entra ID, Microsoft Graph, MSAL (Microsoft Authentication Library), and Managed Identity. Terraform, Docker, and CI/CD (Continuous Integration and Continuous Delivery).

Identity & Security: OAuth 2.0, OIDC (OpenID Connect), SAML 2.0 (Security Assertion Markup Language), JWT (JSON Web Token), JWE (JSON Web Encryption), JWKS (JSON Web Key Set), Firebase Authentication, SSO (Single Sign-On), RBAC (Role-Based Access Control), scoped authorization, envelope encryption, HMAC (Hash-Based Message Authentication Code) blind indexing, and rate limiting.

Data: Microsoft SQL Server, PostgreSQL, Firestore, BigQuery, Redis, Entity Framework Core, LINQ (Language Integrated Query), relational modeling, Code First, migrations, transactions, and cursor-based pagination.

Engineering & Agile Practices: SOLID principles, object-oriented programming, Clean Code, design patterns, dependency injection, refactoring, code review, structured logging, error handling, feature flags, Git, GitHub, Scrum, Azure DevOps, and Jira.

Applied AI: Claude Code, Codex, MCP (Model Context Protocol), skills, subagents, documentation as context, and backlog-to-PR workflow automation. Uses both tools through Visual Studio Code extensions and its integrated terminal to analyze, plan, implement, and review changes with project context and human oversight.

AI Orchestration: Semantic Kernel SDK, LLM Integration, AI Agents, Tool Calling / Function Calling, Agentic Workflows, and Multi-Agent Orchestration.

Knowledge Retrieval: Embeddings, Semantic Search, Vector Search, RAG Architecture, PostgreSQL + pgvector, Cosine Similarity, and Top-K Retrieval.

Model Integration & Inference: ONNX Runtime, Local Model Inference, Hugging Face Tokenization, Embedding Models, Transformer Models, and Model Artifacts.

AI Application Patterns: Human-in-the-Loop, Structured Outputs, Event-Driven Agents, Background Agents, Database Tools, and External API Tools.

AI Provider & SDK Integration: OpenAI SDK, Azure OpenAI SDK, Anthropic SDK, Google Gemini SDK, and REST API Integration.

Integrations: Clever, ClassLink/OneRoster, Google Workspace, Google Admin Directory, Google Classroom API, and Microsoft Graph.

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer & Technical Lead

Private Automotive-Sector Client | Mexico | Remote | 2026

Independent professional project to design and develop an enterprise CRM platform.

- Lead a 3-person team, coordinating a frontend developer and a QA specialist while owning full-stack development, architecture, infrastructure, and client communication.
- Designed the decoupled architecture and lead development using Angular 21, TypeScript, .NET 9, ASP.NET Core Web API, Entity Framework Core 9, and Microsoft SQL Server.
- Implemented modules for customers, vehicles, services, quotations, automotive glass, appointments, users, companies, branches, and configuration, with multi-company and multi-branch context.
- Built an appointment calendar with working schedules, holidays, per-slot capacity, availability rules, and conflict validation.
- Implemented authentication using JWT stored in HttpOnly cookies, global authorization, session control, validation, and centralized exception handling.
- Built responsive administrative interfaces with Angular Material, Signals, reactive forms, and standalone components, plus real-time communication using SSE.
- Manage planning, task creation and assignment, progress reviews, blocker resolution, and technical preparation of the deployment infrastructure.

Senior Full-Stack Software Engineer

Itopia | United States | Remote | 2019-2026

U.S.-based SaaS company serving K-12 education, focused on cloud platforms, virtualization, and remote access.

Technical Leadership & Architecture

- Provided internal technical leadership for a small team comprising one frontend engineer, one backend engineer, and one QA specialist.
- Created and assigned tasks, estimated work, reviewed PRs, guided implementation, and coordinated delivery across frontend, backend, and QA.
- Orchestrated the architecture of IGW (Itopia Gateway), defining domains, integrations, conventions, and technical criteria for its evolution.
- Standardized structured logging, dependency injection, secret management, and code conventions across more than 5 repositories.
- Designed an AI-assisted engineering system using skills, subagents, MCP servers, and documentation as context to automate planning and PR creation.

Next-Generation Multi-Tenant SaaS Platform (2025-2026)

- Designed and implemented microservices for a multi-tenant SaaS platform serving school districts, with a single-entry gateway, REST APIs, OAuth 2.0 client credentials, scoped authorization, and rate limiting.
- Built session, organization, catalog, and internal event-delivery domains using .NET 10, gRPC, and service-to-service OIDC authentication.
- Developed automated tenant onboarding on Azure using state machines, Entra ID consent flows, per-tenant callbacks, event fan-out, and scheduled retries with Cloud Scheduler.
- Designed capacity and cost management for Azure Virtual Desktop host pools using ARM, batch startup, disk SKU adjustments, configurable capacity buffers, and controlled pool growth.
- Implemented long-running asynchronous operations with polling, ghost-session detection and cleanup, feature flags, and multi-layer idempotency.
- Identified and documented authentication vulnerabilities, including audience confusion during token exchange, and designed mitigations through strict validation and HMAC indexes for sensitive data.
- Defined infrastructure as code with Terraform and CI/CD pipelines using Cloud Build.

EdTech VDI Platform (2019-2025)

- Built and evolved a polyglot VDI (Virtual Desktop Infrastructure) platform of approximately 20 C#/.NET and Node.js/TypeScript services deployed on Cloud Run and Cloud Functions.
- Developed rostering and identity integrations with Clever, ClassLink/OneRoster, Google Workspace, Google Admin Directory, Google Classroom API, Microsoft Graph, Entra ID, OAuth 2.0, and SAML 2.0.
- Implemented usage metering and analytics in partitioned BigQuery, virtual-machine orchestration in Compute Engine, and an event engine with Pub/Sub, delay and retry queues, callbacks, and a real-time SSE gateway.
- Built Angular portals for administration, end users, and support, with typed GraphQL code generation, PWA capabilities, and remote-desktop viewer integration.
- Developed a .NET Windows agent for VDI virtual machines, including inactivity detection, event publishing through Pub/Sub, and instance lifecycle management.
- Implemented envelope encryption with Cloud KMS, secret rotation, Terraform infrastructure, and Cloud Build pipelines.
- Developed platform-wide expertise in the WebRTC/GStreamer streaming stack to analyze, diagnose, and evolve it, without claiming authorship of its foundational components.
- Contributed to WorkAnyWhere, Cloud Compliance Manager, CloudApps, and ItopiaOne, products focused on remote connectivity, cloud administration, and consolidated enterprise capabilities.

Project Specialist

XETID | Cuba | 2017-2019

- Developed enterprise systems in the Entity Management Division and contributed to the Distra ARP project.
- Implemented and maintained backend and frontend features using PostgreSQL, Zend Framework, Ext JS, and Bootstrap.
- Analyzed requirements and resolved functional and technical needs within a multidisciplinary team.

Software Engineer - Academic Project

University of Information Sciences (UCI) | Cuba | 2016-2017

- Developed SBRSPPLIT, a system that predicts compatibility between students and courses based on individual aptitudes, as the final degree project.
- Implemented the solution using Python, PyQt5, and PostgreSQL, including Excel file processing with Xlrd, Xlwt, and XlsxWriter, and packaging with PyInstaller.

SELECTED PERSONAL PROJECT

Personal AI Project — Conversational Specialist Search

Designed the conversational workflow and developed retrieval and tool-calling components using Semantic Kernel and RAG over a vectorized specialist catalog.

The workflow design connects a problem description with relevant specialist profiles, uses an LLM to explain matches, and incorporates user clarification to refine retrieval.

EDUCATION

Computer Science Engineering

University of Information Sciences (UCI) | Cuba | 2012-2017

LANGUAGES

- Spanish: Native.
- English: Professional working proficiency in conversations and meetings; advanced reading, writing, and technical vocabulary.